LLM4TSF: Prompt Blocks and Predicted Outputs

Prompts $\Pi^{(k)}$ and evaluator blocks $\Pi_E^{(k)}$ for $k = 0, 1$, instantiated on 29-channel narrowband exposure forecasting.

$k = 0$, Prompt Block $\Pi^{(0)}$ - LLM Reasoning Agent

Task Description $\mathcal{T}$:

You are **LLM4TSF-Reasoner**, an expert in multivariate RF-EMF analysis and cross-channel state-space modelling. Given $\mathbf{X}_{\text{past}}$ and TSFM output θ, produce the state-transition specification $\{\mathbf{F}^{(0)}, \boldsymbol{\delta}_0^{(0)}, \boldsymbol{\rho}_{\text{rel}}^{(0)}\}$ for the Kalman fusion layer.

Objective - generate parameters that:

- Capture cross-channel EMF dependencies via $\mathbf{F} \in \mathbb{R}^{29\times 29}$;
- Provide initial-state corrections $\boldsymbol{\delta}_0 \in \mathbb{R}^{29}$ from boundary dynamics
- Assign reliability weights $\boldsymbol{\rho}_{\text{rel}} \in [0,1]^{29}$: high $\hat{o}_m \Rightarrow$ low $\rho_{\text{rel},m}$

Domain Knowledge:

- 2G (ch 0–7): $\text{autocorr}_{\text{lag-1}} \geq 0.92$; set $f_{ii} \in [0.95, 0.99]$; strong daily periodicity
- 3G–4G cross-operator: moderate working-hour coupling (08:00–20:00 Rome); off-diag $f_{ij} \approx 0.05$–0.12 (e.g. ch 10 TIM – ch 17 VF)
- 5G NR (ch 25–28): near-stationary, low amplitude; $f_{ii} \approx 0.98$, off-diag ≈ 0
- Same-operator intra-band: $f_{ij} \in [0.05, 0.10]$; cross-operator inter-band: $|f_{ij}| \leq 0.05$; high $\hat{o}_{m,h} \Rightarrow$ reduce $\rho_{\text{rel},m}$ toward ϵ_ρ

Environment Details:

- Site: Univ. Hospital Tor Vergata, Rome; July 2023–March 2024; 7.5-min grid
- $M = 29$ channels, 9 kHz–6 GHz; $L = 1344$ steps (7 days); $H = 192$ steps (1 day)
- Operators: Iliad (ch 0–6), TIM (ch 7–13), VF (ch 14–20), W3 (ch 21–28)
- Bands: 2G (ch 0–7), 3G (ch 8–14), 4G (ch 15–24), 5G (ch 25–28)
- Kalman: $R_0 = 2.0$, $q = 10^{-3}$, $\varepsilon_\rho = 0.1$

Historical Context $\mathbf{X}_{\text{past}}$:

`[channel_stats_table]` *(ch_id | operator | band | $\hat{\mu}$ | $\hat{\sigma}$ | trend slope | $\text{autocorr}_{\text{lag-1}}$)*

TSFM Output θ (Chronos-2):

`[tsfm_forecast]`

Few-Shot Examples $\mathcal{S}_M$ (3 nearest from $\mathcal{D}$):

`[retrieved_examples]` *(context summary → $\mathbf{F}$ diag. range → ND/MAPE)*

Instructions:

Think step by step before producing the output:

1. Analyze $\mathbf{X}_{\text{past}}$ per channel: identify trend, periodicity, and $\text{autocorr}_{\text{lag-1}}$
2. Inspect $\boldsymbol{\theta}$: note which channels have large $\hat{o}_{m,h}$ and assign low $\rho_{\text{rel},m}^{(0)}$
3. Use domain knowledge and $\mathcal{S}_M$ to set diagonal $f_{ii}^{(0)}$ per band
4. Identify dominant cross-channel pairs and set off-diagonal $f_{ij}^{(0)}$ accordingly
5. Justify $\boldsymbol{\delta}_0^{(0)}$ from boundary dynamics at $t = 0$
6. Verify all eigenvalues of $\mathbf{F}^{(0)}$ satisfy $|\lambda| \leq 1$ before outputting

Response Format:

1. `<REASONING_TRACE>`: Follow the steps above explicitly; identify dominant channel pairs; justify $\boldsymbol{\delta}_0$ from boundary dynamics;
2. `<OUTPUT>`: Full JSON

Required Output Format:

```
{"F_coefficients":[{"i":int,"j":int,"f_ij":float},...],
 "delta_0":[float x29], "rho_rel":[float x29]}
```

Followed by `<REASONING_TRACE>`.

Output Block of LLM_1 - $k = 0$

Output: $\phi^{(0)} = \{\mathbf{F}^{(0)}, \boldsymbol{\delta}_0^{(0)}, \boldsymbol{\rho}_{\text{rel}}^{(0)}\}$

```
{"F_coefficients":[
   {"i":0, "j":0,  "f_ij":[f_2G_self]},
   {"i":1, "j":0,  "f_ij":[f_2G_intra]},
   {"i":10,"j":15, "f_ij":[f_3G4G_cross]},
   ... (all 29x29 entries) ],
 "delta_0":[[d_ch0],[d_ch1],...,[d_ch28]],
 "rho_rel":[[r_ch0],[r_ch1],...,[r_ch28]]}
```

Reasoning Trace $\text{RT}^{(0)}$:

LLM_1 - Full $\mathbf{F}^{(0)} \in \mathbb{R}^{29\times 29}$ populated as follows. Diagonal: $f_{ii}^{(0)} \in [0.95, 0.99]$ for 2G (ch 0–7) based on high $\text{autocorr}_{\text{lag-1}}$ in $\mathbf{X}_{\text{past}}$; $f_{ii}^{(0)} \in [0.90, 0.96]$ for 3G–4G (ch 8–24); $f_{ii}^{(0)} \approx 0.98$ for near-flat 5G (ch 25–28). Off-diagonal: cross-operator 3G–4G pairs (e.g. ch 10–ch 17, TIM–VF) assigned $f_{ij}^{(0)} \in [0.05, 0.12]$ cross-operator pairs $|f_{ij}^{(0)}| \leq 0.05$; remaining entries near zero. Per-channel $\rho_{\text{rel},m}^{(0)}$: channels with large uncertainty (e.g. ch 17, ch 21) assigned low $\rho_{\text{rel},m}^{(0)} \in [\epsilon_\rho, 0.3]$; channels with small uncertainty (e.g. ch 25–28) assigned $\rho_{\text{rel},m}^{(0)} \in [0.7, 1.0]$. All entries specified.

$k = 0$, Prompt Block $\Pi_E^{(0)}$ - Evaluator Agent

Task Description $\mathcal{T}_E$:

You are **LLM4TSF-Evaluator**. Analyze the submitted state-space parameters for plausibility and consistency with $\mathbf{X}_{\text{past}}$ and TSFM uncertainty $\hat{\mathbf{o}}$. Do **not** produce a forecast. Output *only* $\Delta^{(1)}$: *"No major changes needed"* or at most 5 lines of targeted revision instructions.

Environment Details:

(Same as Reasoning Agent) $M = 29$; Iliad/TIM/VF/W3; 2G/3G/4G/5G; $L = 1344$, $H = 192$, $R_0 = 2.0$, $q = 10^{-3}$

Domain Knowledge:

(Same as Reasoning Agent)

TSFM Output θ:

`[tsfm_forecast]`

Historical Context $\mathbf{X}_{\text{past}}$:

`[channel_stats_table]`

LLM Reasoning Trace $\text{RT}^{(0)}$:

`[reasoning_trace_k0]`

Submitted $\phi^{(0)}$:

$\mathbf{F}$: `[F_submitted]` $\boldsymbol{\delta}_0$: `[delta0_sub]` $\boldsymbol{\rho}_{\text{rel}}$: `[rho_sub]`

Few-Shot Examples $\mathcal{S}_E$:

`[eval_examples]` *(F submitted → issue detected → Δ given → MAPE gain)*

Chain-of-Thought Evaluation Instructions:

Think step by step before producing feedback:

1. Check diagonal entries $f_{ii}^{(0)}$ against band-level $\text{autocorr}_{\text{lag-1}}$ in $\mathbf{X}_{\text{past}}$; flag any that violate domain knowledge ranges
2. Verify off-diagonal entries $f_{ij}^{(0)}$ are consistent with operator and band coupling rules; flag entries outside expected ranges
3. Assess $\boldsymbol{\rho}_{\text{rel}}^{(0)}$–$\hat{o}_{m,h}$ inverse alignment: channels with large QS_m should have low $\rho_{\text{rel},m}^{(0)}$; flag violations
4. Check $\boldsymbol{\delta}_0^{(0)}$ is appropriately scaled relative to historical channel variations in $\mathbf{X}_{\text{past}}$
5. Inspect $\text{RT}^{(0)}$ for consistency between stated reasoning and submitted values
6. Summarize only flagged issues as $\Delta^{(1)}$; if none, output *"No major changes needed"*

Response Format:

1. `<FEEDBACK>`: Output $\Delta^{(1)}$ only

Output Block of LLM_E - $k = 0$

Output Feedback:
$\Delta^{(1)}$: "Overall $\mathbf{F}^{(0)} \in \mathbb{R}^{29\times29}$ structure is plausible. Two per-channel issues detected. (1) Off-diagonal $f^{(0)}_{10,17}$ (ch 10–ch 17, 3G–4G, TIM) exceeds the expected range $[0.05, 0.12]$ for moderate cross-band coupling – reduce to within $[0.05, 0.12]$. (2) $\rho^{(0)}_{\text{rel},17}$ for ch 17 is too high, reduce $\rho^{(0)}_{\text{rel},17}$ to shift Kalman trust toward LLM-defined dynamics for this channel. All other entries consistent."

$k = 1$, Prompt Block $\Pi^{(1)}$ - LLM Reasoning Agent

Evaluator Feedback $\Delta^{(1)}$ [*address every point first*]:
`[delta_1_full]`

Previous Output $\phi^{(0)}$ and History:
$\mathcal{H}^{(1)} = \mathcal{H}^{(0)} \cup \{\Delta^{(1)}, \phi^{(0)}, \text{RT}^{(0)}\}$; $\mathcal{H}^{(0)} = \{\mathbf{X}_{\text{past}}, \theta, \mathcal{S}_M, \mathcal{T}\}$
$\mathbf{F}$: `[F_k0]` $\boldsymbol{\delta}_0$: `[d0k0]` $\boldsymbol{\rho}_{\text{rel}}$: `[rhok0]`

Previous Reasoning Trace $\text{RT}^{(0)}$:
`[reasoning_trace_k0]`

Response Format:

1. `<REASONING_TRACE>`: For each evaluator flag - state what was wrong, how it is corrected, and why the correction is consistent with EMF domain knowledge and θ.
2. `<OUTPUT>`: Full revised JSON (all 29×29 entries; do not omit unchanged entries)

Required Output Format:

```
{"F_coefficients":[{"i":int,"j":int,"f_ij":float},...],
 "delta_0":[float x29], "rho_rel":[float x29]}
```

Output Block of LLM_1 - $k = 1$

Output: $\phi^{(1)} = \{\mathbf{F}^{(1)}, \boldsymbol{\delta}^{(1)}_0, \boldsymbol{\rho}^{(1)}_{\text{rel}}\}$

```
{"F_coefficients":[
   {"i":0, "j":0,  "f_ij":[f_2G_self]},       % unchanged
   {"i":10,"j":17, "f_ij":[f_3G4G_revised]},   % corrected
   ... (all 29x29 entries) ],
 "delta_0":[[d_ch0],...,[d_ch28]],              % unchanged
 "rho_rel":[[r_ch0],...,[r_revised],...,[r_ch28]]}
```

Reasoning Trace $\text{RT}^{(1)}$:
LLM_1 - Applied $\Delta^{(1)}$ to $\mathbf{F}^{(0)} \in \mathbb{R}^{29\times29}$: (1) reduced $f^{(1)}_{10,17}$ to within $[0.05, 0.12]$, consistent with moderate 3G–4G cross-operator coupling (ch 10, TIM; ch 17, VF) observed in $\mathbf{X}_{\text{past}}$. (2) reduced $\rho^{(1)}_{\text{rel},17}$ toward ϵ_ρ, consistent with ch 17 exhibiting one of the largest uncertainty values across all 29 channels. All remaining entries of $\mathbf{F}^{(1)}$, $\boldsymbol{\delta}^{(1)}_0$, and $\boldsymbol{\rho}^{(1)}_{\text{rel}}$ carried unchanged from $\boldsymbol{\phi}^{(0)}$.

$k = 1$, Prompt Block $\Pi^{(1)}_E$ - Evaluator Agent

This is the evaluation round $k = 1$. Determine whether $\phi^{(1)}$ correctly incorporates your prior feedback $\Delta^{(1)}$. If all issues are resolved output *"No major changes needed. Proceed to Kalman fusion."* Otherwise output residual corrections as $\Delta^{(2)}$ (max 5 lines).

Required Output Format:
$\Delta^{(2)}$: `<feedback or PASS>`

Prior Feedback $\Delta^{(1)}$:
`[delta_1_full]`

Revised Submission $\phi^{(1)}$ (under review):
$\mathbf{F}$: `[Fk1]` $\boldsymbol{\delta}_0$: `[d0k1]` $\boldsymbol{\rho}_{\text{rel}}$: `[rhok1]`

LLM Reasoning Trace $\text{RT}^{(1)}$:
`[reasoning_trace_k1]`

Response Format:

1. `<FEEDBACK>`: Output $\Delta^{(2)}$ only

Output Block of LLM_E - $k = 1$

Output Feedback:
$\Delta^{(2)}$: "No major changes needed. Proceed to Kalman fusion. Revised $f^{(1)}_{10,17}$ now within $[0.05, 0.12]$, consistent with 3G–4G working-hour coupling in $\mathbf{X}_{\text{past}}$ (TIM, ch 10 and ch 17). Updated $\rho^{(1)}_{\text{rel},17}$ toward ϵ_ρ correctly reflects the large uncertainty for ch 17. All entries of $\mathbf{F}^{(1)}$ verified; all $\boldsymbol{\delta}^{(1)}_0$ entries consistent with $\mathbf{X}_{\text{past}}$ boundary conditions. $\boldsymbol{\phi}^{(1)}$ accepted."

LLM4TSF: Prompt Configurations for Prompt-Sensitivity Analysis

Three prompt used in Table IV of the main paper, **Highlighted sections** indicate additions over the previous prompt.

Prompt 1 - Task Description Only

Task Description $\mathcal{T}$:

You are **LLM4TSF-Reasoner**, an expert in multivariate RF-EMF analysis and cross-channel state-space modelling. Given $\mathbf{X}_{\text{past}}$ and TSFM output $\boldsymbol{\theta}$, produce the state-transition specification $\{\mathbf{F}^{(0)}, \boldsymbol{\delta}_0^{(0)}, \boldsymbol{\rho}_{\text{rel}}^{(0)}\}$ for the Kalman fusion layer.

Objective - generate parameters that:

- Capture cross-channel EMF dependencies via $\mathbf{F} \in \mathbb{R}^{29\times 29}$;
- Provide initial-state corrections $\boldsymbol{\delta}_0 \in \mathbb{R}^{29}$ from boundary dynamics
- Assign reliability weights $\boldsymbol{\rho}_{\text{rel}} \in [0,1]^{29}$: high $\hat{o}_{m,h} \Rightarrow$ low $\rho_{\text{rel},m}$

Domain Knowledge:

- 2G (ch 0–7): autocorr$_{\text{lag-1}} \geq 0.92$; set $f_{ii} \in [0.95, 0.99]$; strong daily periodicity
- 3G–4G cross-operator: moderate working-hour coupling (08:00–20:00 Rome); off-diag $f_{ij} \approx 0.05$–0.12 (e.g. ch 10 TIM – ch 17 VF)
- 5G NR (ch 25–28): near-stationary, low amplitude; $f_{ii} \approx 0.98$, off-diag ≈ 0
- Same-operator intra-band: $f_{ij} \in [0.05, 0.10]$; cross-operator inter-band: $|f_{ij}| \leq 0.05$; high $\hat{o}_{m,h} \Rightarrow$ reduce $\rho_{\text{rel},m}$ toward ϵ_ρ

Environment Details:

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- Bands: 2G (ch 0–7), 3G (ch 8–14), 4G (ch 15–24), 5G (ch 25–28)
- Kalman: $R_0 = 2.0$, $q = 10^{-3}$, $\varepsilon_\rho = 0.1$

Historical Context $\mathbf{X}_{\text{past}}$:

`[channel_stats_table]` *(ch_id | operator | band | $\hat{\mu}$ | $\hat{\sigma}$ | trend slope | autocorr$_{\text{lag-1}}$)*

TSFM Output $\boldsymbol{\theta}$ (Chronos-2):

`[tsfm_forecast]`

Response Format:

1. `<REASONING_TRACE>`: Identify dominant channel pairs; justify $\boldsymbol{\delta}_0$ from boundary dynamics;
2. `<OUTPUT>`: Full JSON

Required Output Format:

```
{"F_coefficients":[{"i":int,"j":int,"f_ij":float},...],
 "delta_0":[float x29], "rho_rel":[float x29]}
```

Followed by `<REASONING_TRACE>`.

Prompt 2 - Task Description + Few-Shot Examples $\mathcal{S}_M$

Task Description $\mathcal{T}$:

You are **LLM4TSF-Reasoner**, an expert in multivariate RF-EMF analysis and cross-channel state-space modelling. Given $\mathbf{X}_{\text{past}}$ and TSFM output $\boldsymbol{\theta}$, produce the state-transition specification $\{\mathbf{F}^{(0)}, \boldsymbol{\delta}_0^{(0)}, \boldsymbol{\rho}_{\text{rel}}^{(0)}\}$ for the Kalman fusion layer.

Objective - generate parameters that:

- Capture cross-channel EMF dependencies via $\mathbf{F} \in \mathbb{R}^{29\times 29}$;
- Provide initial-state corrections $\boldsymbol{\delta}_0 \in \mathbb{R}^{29}$ from boundary dynamics
- Assign reliability weights $\boldsymbol{\rho}_{\text{rel}} \in [0,1]^{29}$: high $\hat{o}_{m,h} \Rightarrow$ low $\rho_{\text{rel},m}$

Environment Details:

- Site: Univ. Hospital Tor Vergata, Rome; July 2023–March 2024; 7.5-min grid
- $M = 29$ channels, 9 kHz–6 GHz; $L = 1344$ steps (7 days); $H = 192$ steps (1 day)
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- Bands: 2G (ch 0–7), 3G (ch 8–14), 4G (ch 15–24), 5G (ch 25–28)
- Kalman: $R_0 = 2.0$, $q = 10^{-3}$, $\varepsilon_\rho = 0.1$

Historical Context $\mathbf{X}_{\text{past}}$:

`[channel_stats_table]` *(ch_id | operator | band | $\hat{\mu}$ | $\hat{\sigma}$ | trend slope | autocorr$_{\text{lag-1}}$)*

TSFM Output $\boldsymbol{\theta}$ (Chronos-2):

`[tsfm_forecast]`

$\oplus$ Few-Shot Examples $\mathcal{S}_M$ (3 nearest from $\mathcal{D}$):

`[retrieved_examples]` *(context summary → $\mathbf{F}$ diag. range → ND/MAPE)*

Response Format:

1. `<REASONING_TRACE>`: Identify dominant channel pairs; justify $\boldsymbol{\delta}_0$ from boundary dynamics;
2. `<OUTPUT>`: Full JSON

Required Output Format:

```
{"F_coefficients":[{"i":int,"j":int,"f_ij":float},...],
 "delta_0":[float x29], "rho_rel":[float x29]}
```

Followed by `<REASONING_TRACE>`.

Prompt 3 - Task Description + Few-Shot Examples $\mathcal{S}_M$ + Chain-of-Thought

Task Description $\mathcal{T}$:

You are **LLM4TSF-Reasoner**, an expert in multivariate RF-EMF analysis and cross-channel state-space modelling. Given $\mathbf{X}_{\text{past}}$ and TSFM output $\boldsymbol{\theta}$, produce the state-transition specification $\{\mathbf{F}^{(0)}, \boldsymbol{\delta}_0^{(0)}, \boldsymbol{\rho}_{\text{rel}}^{(0)}\}$ for the Kalman fusion layer.

Objective - generate parameters that:

- Capture cross-channel EMF dependencies via $\mathbf{F} \in \mathbb{R}^{29\times 29}$;
- Provide initial-state corrections $\boldsymbol{\delta}_0 \in \mathbb{R}^{29}$ from boundary dynamics
- Assign reliability weights $\boldsymbol{\rho}_{\text{rel}} \in [0,1]^{29}$: high $\hat{o}_{m,h} \Rightarrow$ low $\rho_{\text{rel},m}$

Environment Details:

- Site: Univ. Hospital Tor Vergata, Rome; July 2023–March 2024; 7.5-min grid
- $M = 29$ channels, 9 kHz–6 GHz; $L = 1344$ steps (7 days); $H = 192$ steps (1 day)
- Operators: Iliad (ch 0–6), TIM (ch 7–13), VF (ch 14–20), W3 (ch 21–28)
- Bands: 2G (ch 0–7), 3G (ch 8–14), 4G (ch 15–24), 5G (ch 25–28)
- Kalman: $R_0 = 2.0$, $q = 10^{-3}$, $\varepsilon_\rho = 0.1$

Historical Context $\mathbf{X}_{\text{past}}$:

`[channel_stats_table]` *(ch_id | operator | band | $\hat{\mu}$ | $\hat{\sigma}$ | trend slope | autocorr$_{\text{lag-1}}$)*

TSFM Output $\boldsymbol{\theta}$ (Chronos-2):

`[tsfm_forecast]`

Few-Shot Examples $\mathcal{S}_M$ (3 nearest from $\mathcal{D}$):

`[retrieved_examples]` *(context summary → $\mathbf{F}$ diag. range → ND/MAPE)*

$\oplus$ Chain-of-Thought Reasoning Instructions:

Think step by step before producing the output:

1. Analyze $\mathbf{X}_{\text{past}}$ per channel: identify trend, periodicity, and autocorr$_{\text{lag-1}}$
2. Inspect $\boldsymbol{\theta}$: note which channels have large $\hat{o}_{m,h}$ and assign low $\rho_{\text{rel},m}^{(0)}$
3. Use domain knowledge and $\mathcal{S}_M$ to set diagonal $f_{ii}^{(0)}$ per band
4. Identify dominant cross-channel pairs and set off-diagonal $f_{ij}^{(0)}$ accordingly
5. Justify $\boldsymbol{\delta}_0^{(0)}$ from boundary dynamics at $t = 0$
6. Verify all eigenvalues of $\mathbf{F}^{(0)}$ satisfy $|\lambda| \leq 1$ before outputting

Response Format:

1. `<REASONING_TRACE>`: Follow the steps above explicitly; identify dominant channel pairs; justify $\boldsymbol{\delta}_0$ from boundary dynamics;
2. `<OUTPUT>`: Full JSON

Required Output Format:

```
{"F_coefficients":[{"i":int,"j":int,"f_ij":float},...],
 "delta_0":[float x29], "rho_rel":[float x29]}
```

Followed by `<REASONING_TRACE>`.